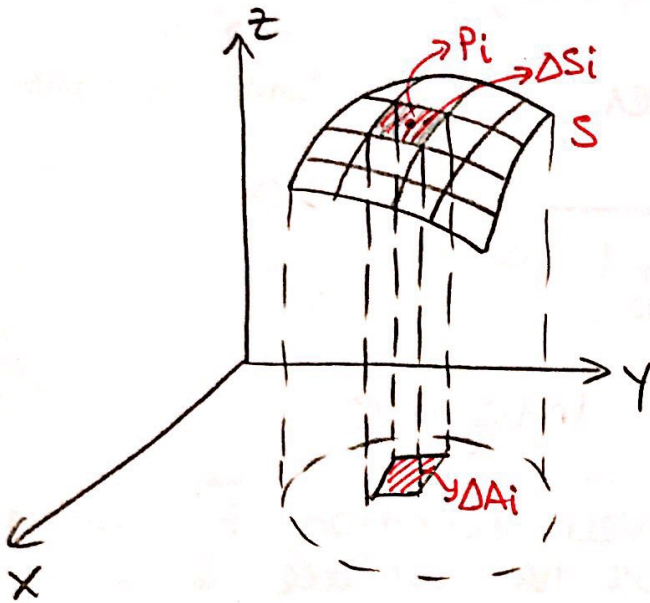


SURFACE INTEGRALS



Let consider the surface defined by $z=f(x,y)$ - let have the partition of S and denote surface elements by ΔS_i .

Now consider the function $G(x,y,z)$ which is defined and continuous on the points of this surface.

Choose arbitrary points $P_i(x_i, y_i, z_i)$ in each ΔS_i and evaluate $G(P_i)$.

$$\sum_{i=1}^n G(P_i) \Delta S_i = \sum_{i=1}^n G(x_i, y_i, z_i) \cdot \Delta S_i$$

Let $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n G(x_i, y_i, z_i) \Delta S_i = \iint_S G(x, y, z) ds \quad \text{surface}$$

integral of the function $G(x, y, z)$ on the surface S.

If the partition of surface S is made by the parallel planes then the projection of each ΔS_i will be rectangular regions.

⊗ If $G(x, y, z) \equiv 1$ then $G(P_i) = 1$, so

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta S_i = \iint_S ds \quad \text{gives the area of the surface S.}$$

Now we have to find Δs_i ;

let θ_i be the angle between z -axis and $\vec{n}_i$ (normal at the points P_i);

$$S = \iint_S G(x, y, z) \cdot \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dx dy$$

Vector Form of Surface Integrals

Let consider the vector valued function $\vec{F}$ which is defined and continuous on the surface S .

$$\vec{F} = L(x, y, z)\vec{i} + M(x, y, z)\vec{j} + N(x, y, z)\vec{k}$$

Then consider the unit normal vector $\vec{n}$ of S . Denote the direction cosine of normal vector by $\cos \alpha, \cos \beta, \cos \gamma$. Then;

$$\vec{n} = \cos \alpha \vec{i} + \cos \beta \vec{j} + \cos \gamma \vec{k}$$

$$\vec{n} = \cos(\underbrace{n, x}_{\alpha})\vec{i} + \cos(\underbrace{n, y}_{\beta})\vec{j} + \cos(\underbrace{n, z}_{\gamma})\vec{k} \quad \text{and}$$

$$\sum_{i=1}^n \vec{F}(P_i) \cdot \vec{n}_i \Delta s_i \quad \text{and let } n \rightarrow \infty \text{ for } \Delta s_i \rightarrow 0,$$
$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \vec{F}(P_i) \cdot \vec{n}_i \Delta s_i = \iint_S \vec{F} \cdot \vec{n} ds \quad \text{is a surface integral on } S$$

$$\iint_S \vec{F} \cdot \vec{n} ds = \iint_S (L \cos \alpha + M \cos \beta + N \cos \gamma) ds$$

$$* \cos \alpha ds = dy dz \quad * \cos \beta ds = dx dz \quad * \cos \gamma ds = dx dy$$

such that,

$$(*) ds = \sec \gamma dx dy, \quad (*) ds = \sec \beta dx dz \quad (*) ds = \sec \alpha dy dz$$

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$\Delta s \cdot \cos(n, z)$ gives the projection of Δs on xy -plane,
 $\Delta s \cdot \cos(n, x)$ " " " Δs on yz -plane,
 $\Delta s \cdot \cos(n, y)$ " " " Δs on xz -plane.

Therefore;

$$\iint_S \vec{F} \cdot \vec{n} ds = \iint_S L dydz + M dx dz + N dx dy$$

Evaluation of Surface Integrals

The surface S is given by $z = f(x, y)$ and $\vec{n}$ is the unit normal vector of f .

$$z = f(x, y) \Rightarrow z - f(x, y) = 0.$$

$$\vec{n} = \frac{-\frac{\partial z}{\partial x} \vec{i} - \frac{\partial z}{\partial y} \vec{j} + \vec{k}}{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}}$$

$$\vec{n} \cdot \vec{k} = \frac{1}{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}} \Rightarrow \sec \gamma = \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}$$

Since

$$ds = \sec \gamma dx dy;$$

$$\iint_S \vec{F} \cdot \vec{n} ds = \iint_S (L\vec{i} + M\vec{j} + N\vec{k}) \left(\frac{-\frac{\partial z}{\partial x} \vec{i} - \frac{\partial z}{\partial y} \vec{j} + \vec{k}}{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}} \right) \cdot \frac{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}}{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}} dx dy$$

$$\iint_S \vec{F} \cdot \vec{n} ds = \iint_{D_1} \left(-L \frac{\partial z}{\partial x} - M \frac{\partial z}{\partial y} + N \right) dx dy$$

projection
onto
 xy -plane

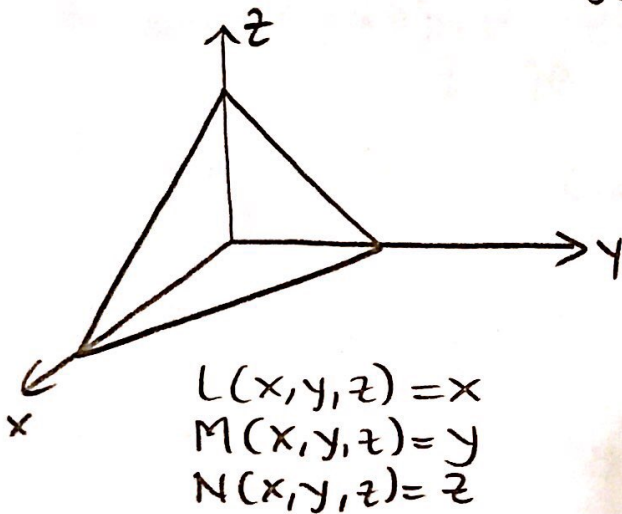
$$\iint_S \vec{F} \cdot \vec{n} ds = \iint_{D_2} \left(L - M \frac{\partial x}{\partial y} - N \frac{\partial x}{\partial z} \right) dy dz$$

projection onto
yz-plane

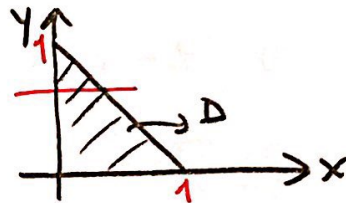
$$\iint_S \vec{F} \cdot \vec{n} ds = \iint_{D_3} \left(-L \frac{\partial y}{\partial x} + M - N \frac{\partial y}{\partial z} \right) dx dz$$

projection
onto xz-plane

Example! Surface S is given by $x+y+z=1$
then evaluate $I = \iint_S x dy dz + y dz dx + z dx dy$.



projection onto xy-plane



$$* z = 1 - x - y$$

$$\iint_S \vec{F} \cdot \vec{n} ds = \iint_D \left(-L \frac{\partial z}{\partial x} - M \frac{\partial z}{\partial y} + N \right) dx dy$$

$$\iint_S x dy dz + y dz dx + z dx dy = \iint_D [-x \cdot (-1) - y \cdot (-1) + z] dx dy$$

$$I = \iint_D (x + y + z) dx dy$$

$$I = \int_0^1 \int_0^{1-y} (\cancel{x} + y + \overbrace{1-x-y}^z) dx dy = \int_0^1 x \Big|_0^{1-y} dy = \int_0^1 (1-y) dy$$

$$I = \left[y - \frac{y^2}{2} \right]_0^1 = 1 - \frac{1}{2} = \frac{1}{2} //$$

2nd way: $\vec{F} = L\vec{i} + M\vec{j} + N\vec{k}$

$$\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$x+y+z-1=0 \Rightarrow \vec{n} = \frac{\vec{i} + \vec{j} + \vec{k}}{\sqrt{3}}$$

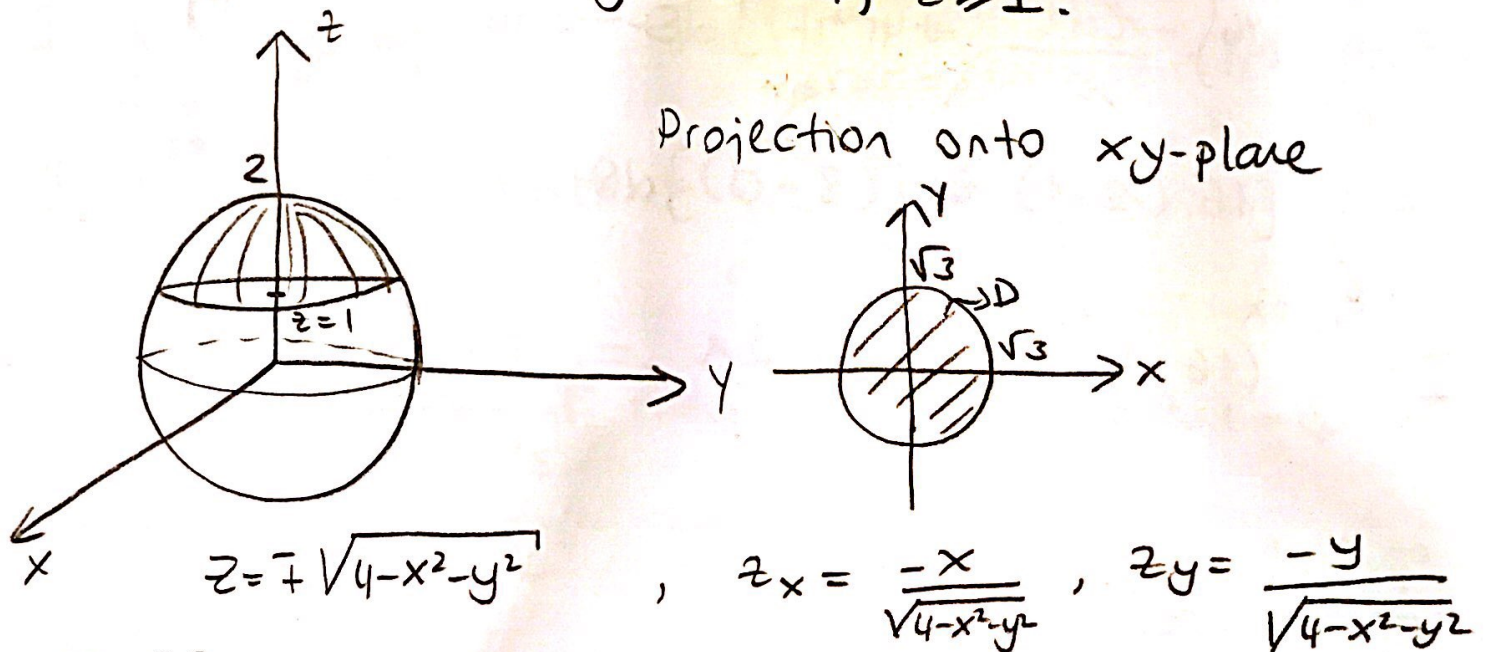
$$\vec{n} \cdot \vec{k} = \cos \delta = \frac{1}{\sqrt{3}} \Rightarrow \sec \delta = \sqrt{3}$$

$$ds = \sec \delta dx dy \Rightarrow ds = \sqrt{3} dx dy$$

$$I = \iint_S \vec{F} \cdot \vec{n} ds = \iint_D (x+y+z) \cdot \frac{1}{\sqrt{3}} \cdot \sqrt{3} dx dy$$

$$I = \iint_D (x+y+1-x-y) dx dy = \iint_D dx dy = \frac{1}{2}$$

Example: Evaluate $I = \iint_S [x^2 + y^2 + (z-2)^2] ds$ on the surface $S: x^2 + y^2 + z^2 = 4, z \geq 1$.



$$I = \iint_S [x^2 + y^2 + (z-2)^2] \cdot \sec \delta dx dy$$

$$\sec \delta = \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} = \sqrt{\frac{x^2 + y^2}{4 - x^2 - y^2} + 1} = \frac{2}{\sqrt{4 - x^2 - y^2}}$$

$$I = \iint_D [x^2 + y^2 + \underbrace{(z-2)}_{z = \sqrt{4-x^2-y^2}}]^2 \cdot \frac{2}{\sqrt{4-x^2-y^2}} dx dy$$

$$I = \iint_D (\cancel{x^2+y^2} + 4 - \cancel{x^2-y^2} - 4 \cdot \sqrt{4-x^2-y^2} + 4) \cdot \frac{2}{\sqrt{4-x^2-y^2}} dx dy$$

$$I = \iint_D (8 - 4 \cdot \sqrt{4-x^2-y^2}) \frac{2}{\sqrt{4-x^2-y^2}} dx dy$$

$$I = \iint_D \left(\frac{16}{\sqrt{4-x^2-y^2}} - 8 \right) dx dy, \quad \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$I = \int_0^{2\pi} \int_0^{\sqrt{3}} \left(\frac{16}{\sqrt{4-r^2}} - 8 \right) r dr d\theta = \int_0^{2\pi} \left(\frac{16r}{\sqrt{4-r^2}} - 8r \right) d\theta$$

$4-r^2 = t^2$
 $-2r dr = 2t dt$

$$I = \int_0^{2\pi} \left[16 \int_2^1 \frac{-t dt}{t} - 4r^2 \Big|_0^{\sqrt{3}} \right] d\theta$$

$$I = \int_0^{2\pi} [16 \cdot (\underline{2-1}) - 4(3-0)] d\theta$$

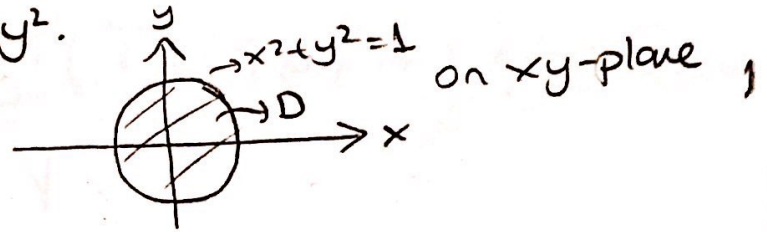
$$I = \int_0^{2\pi} (16-12) d\theta = 4 \cdot 2\pi \Rightarrow \underline{\underline{I = 8\pi}}$$

Example: Evaluate $I = \iint_S z dx dy + x dy dz + y dx dz$ such that $S: z = \sqrt{1-x^2-y^2}$. [8]

such that $S: z = \sqrt{1-x^2-y^2}$.

$$\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$$

$\downarrow$ $\downarrow$ $\downarrow$
 L M N



$$\iint_S \vec{F} \cdot \vec{n} ds = \iint_D (-L \frac{\partial z}{\partial x} - M \frac{\partial z}{\partial y} + N) dx dy$$

$$I = \iint_D \left[-x \cdot \left(\frac{-x}{\sqrt{1-x^2-y^2}} \right) - y \cdot \left(\frac{-y}{\sqrt{1-x^2-y^2}} \right) + z \right] dx dy$$

$$I = \iint_D \left[\frac{x^2 + y^2}{\sqrt{1-x^2-y^2}} + \sqrt{1-x^2-y^2} \right] dx dy = \iint_D \frac{x^2 + y^2 + 1 - x^2 - y^2}{\sqrt{1-x^2-y^2}} dx dy$$

$$I = \iint_D \frac{1}{\sqrt{1-x^2-y^2}} dx dy, \text{ polar coord. } \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$I = \int_0^{2\pi} \int_0^1 \frac{1}{\sqrt{1-r^2}} r dr d\theta, \quad \begin{aligned} 1-r^2 &= t^2 \\ -2r dr &= 2t dt \Rightarrow r dr = -t dt \end{aligned}$$

$$I = \int_0^{2\pi} \left(\int_1^0 -\frac{t dt}{t} \right) d\theta = \int_0^{2\pi} (t \Big|_0^1) d\theta = 2\pi //$$

2nd way: $I = \iint_S \vec{F} \cdot \vec{n} ds$, $x^2 + y^2 + z^2 - 1 = 0$

$$\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}, \quad \vec{n} = \text{gradye} = \frac{2x\vec{i} + 2y\vec{j} + 2z\vec{k}}{\sqrt{4x^2 + 4y^2 + 4z^2}}$$

$$\vec{n} \cdot \vec{k} = \frac{2z}{\sqrt{4x^2 + 4y^2 + 4z^2}} = \cos \gamma$$

$$\Rightarrow \sec \gamma = \frac{\sqrt{4x^2 + 4y^2 + 4z^2}}{2z}, \quad ds = \sec \gamma dx dy$$

$$I = \iint_S \vec{F} \cdot \vec{n} \, ds$$

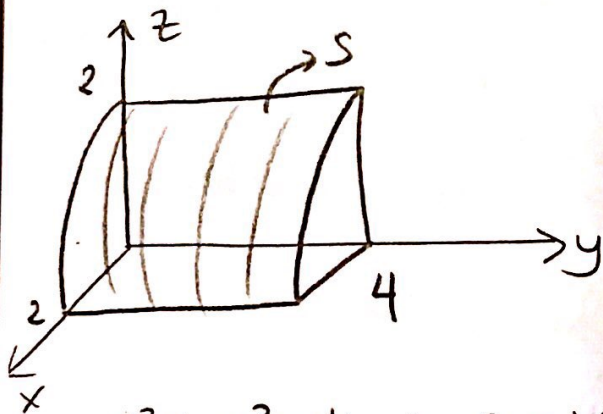
$$I = \iint_D (x\vec{i} + y\vec{j} + z\vec{k}) \cdot \frac{(2x\vec{i} + 2y\vec{j} + 2z\vec{k})}{\sqrt{4x^2 + 4y^2 + 4z^2}} \cdot \frac{\sqrt{4x^2 + 4y^2 + 4z^2}}{2z} \, dx \, dy$$

$$I = \iint_D \frac{(2x^2 + 2y^2 + 2z^2)}{2\sqrt{1-x^2-y^2}} \, dx \, dy = \iint_D \frac{(x^2 + y^2 + 1 - x^2 - y^2)}{\sqrt{1-x^2-y^2}} \, dx \, dy$$

$$I = \iint_D \frac{1}{\sqrt{1-x^2-y^2}} \, dx \, dy \dots \Rightarrow I = 2\pi$$

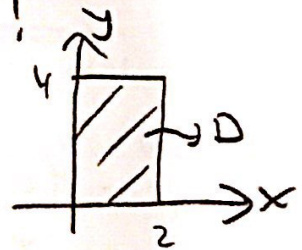
Example: Evaluate the surface integral for

$\vec{F} = xy\vec{i} - z\vec{j} + yz\vec{k}$ on $S: \begin{cases} x^2 + z^2 = 4, x=0, y=0 \\ z=0, y=4 \end{cases}$ in the 1st quad.



⊗ We can not have projection onto xz-plane!

projection onto xy-plane:



$$x^2 + z^2 = 4 \Rightarrow z = \sqrt{4-x^2}$$

$$\vec{F} = \overset{L}{xy}\vec{i} - \overset{M}{z}\vec{j} + \overset{N}{yz}\vec{k}$$

$$I = \iint_S \vec{F} \cdot \vec{n} \, ds = \iint_S xy \, dy \, dz - z \, dx \, dz + yz \, dx \, dy$$

$$I = \iint_D \left(-L \cdot \frac{\partial z}{\partial x} - M \frac{\partial z}{\partial y} + N \right) dx \, dy$$

$$I = \iint_D \left[-xy \cdot \frac{(-x)}{\sqrt{4-x^2}} + z \cdot 0 + yz \right] dx \, dy$$

$$I = \iint_D \left[\frac{x^2 y}{\sqrt{4-x^2}} + y \cdot \sqrt{4-x^2} \right] dx dy$$

$$I = \iint_D \left(\frac{x^2 y + y(4-x^2)}{\sqrt{4-x^2}} \right) dx dy = \iint_D \left(\frac{\cancel{x^2 y} + 4y - \cancel{x^2 y}}{\sqrt{4-x^2}} \right) dx dy$$

$$I = \iint_D \frac{4y}{\sqrt{4-x^2}} dx dy = \int_0^2 \int_0^4 \frac{4y}{\sqrt{4-x^2}} dy dx$$

$$I = \int_0^2 \frac{1}{\sqrt{4-x^2}} \cdot 2y^2 \Big|_0^4 dx = \int_0^2 \frac{2}{\sqrt{4-x^2}} [16-0] dx$$

$$I = 32 \int_0^2 \frac{dx}{\sqrt{4-x^2}} = 32 \cdot \int_0^2 \frac{dx}{2\sqrt{1-(\frac{x}{2})^2}} = 32 \int_0^2 \frac{\frac{1}{2}}{\sqrt{1-(\frac{x}{2})^2}} dx$$

$$I = 32 \cdot \text{Arcsin}\left(\frac{x}{2}\right) \Big|_0^2 = 32 \cdot \left[\frac{\pi}{2} - 0 \right]$$

$$I = 16\pi$$

2nd way: $\vec{F} = xy\vec{i} - z\vec{j} + yz\vec{k}$

$$z = \sqrt{4-x^2} \Rightarrow x^2 + z^2 - 4 = 0$$

$$\vec{n} = \frac{2x\vec{i} + 0\vec{j} + 2z\vec{k}}{\sqrt{4x^2 + 4z^2}}, \quad \vec{n} \cdot \vec{k} = \cos \gamma$$

$$\cos \gamma = \frac{2z}{\sqrt{4x^2 + 4z^2}} \Rightarrow \sec \gamma = \frac{\sqrt{4x^2 + 4z^2}}{2z}$$

$$I = \iint_S \vec{F} \cdot \vec{n} ds = \iint_D \left[xy\vec{i} - z\vec{j} + yz\vec{k} \right] \cdot \frac{2x\vec{i} + 2z\vec{k}}{\sqrt{4x^2 + 4z^2}} \cdot \frac{\sqrt{4x^2 + 4z^2}}{2z} dx dy$$

$$I = \iint_D \frac{(2x^2 y + 2y z^2)}{2z} dx dy$$

$$I = \iint_D \frac{\cancel{2}x^2y + \cancel{2}y(4-x^2)}{\cancel{2}\sqrt{4-x^2}} dx dy$$

$$I = \iint_D \frac{\cancel{x^2}y + 4y - \cancel{x^2}y}{\sqrt{4-x^2}} dx dy$$

$$I = \iint_D \frac{4y}{\sqrt{4-x^2}} dx dy = 16\pi$$

Stoke's Theorem

In this theorem, a line integral in space is transformed to a surface integral. Let the surface S is an open surface which is enclosed by the curve λ . For any point on the surface S , if the angle between normal line and coordinate axes are α, β, γ then;

$$\vec{n} = \cos \alpha \vec{i} + \cos \beta \vec{j} + \cos \gamma \vec{k}$$

$$\vec{n} = \frac{-\frac{\partial z}{\partial x} \vec{i} - \frac{\partial z}{\partial y} \vec{j} + \vec{k}}{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}}$$

Let $\vec{n}$ be the acute angle of positive normal of the surface and z -axis.

